

ANALYTIC HAHN-ECHO DEPHASING FUNCTIONAL

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ABSTRACT. This supplement derives the OU-averaged Hahn-echo attenuation factor used in the numerical simulation. Although the closed-form expression is not used explicitly in the main text, it motivates the stochastic phase accumulation model and provides a consistency check for the Monte Carlo simulation.

We wish to justify the use of the following expression for $S \equiv S(2\tau)$ used in `ou.py` assuming no T_1 correction, namely:

```
def ou_hahn_echo_analytic(tau_values, T1, T2, omega0,
    gamma=2.675e8, T1_correction=False):

    ...

    I = (
        4 * tau_c * tau
        - 6 * tau_c**2
        + 8 * tau_c**2 * np.exp(-tau / tau_c)
        - 2 * tau_c**2 * np.exp(-2 * tau / tau_c)
    )

    if T1_correction:
        S = np.exp(-0.5 * gamma**2 * Delta2 * I)
        * np.exp(-tau / T1)
    else:
        S = np.exp(-0.5 * gamma**2 * Delta2 * I)

    ...
```

In other words, we want to show that

$$S(2\tau) = \exp \left[-\frac{1}{2} \gamma^2 \Delta^2 (4\tau_c \tau - 6\tau_c^2 + 8\tau_c^2 e^{-\tau/\tau_c} - 2\tau_c^2 e^{-2\tau/\tau_c}) \right]$$

1. FIELD MODEL

We define the Ornstein-Uhlenbeck fluctuating magnetic field $\mathfrak{B}(t)$ satisfying

$$\langle \mathfrak{B}(t) \rangle = 0, \quad \langle \mathfrak{B}(t) \mathfrak{B}(t + \tau) \rangle = \Delta^2 e^{-|\tau|/\tau_c}.$$

The accumulated phase is therefore

$$\phi(2\tau) = \gamma \int_0^{2\tau} y(t) \mathfrak{B}(t) dt,$$

where the Hahn-echo modulation function is given by

$$y(t) = \begin{cases} +1, & 0 < t < \tau \\ -1, & \tau < t < 2\tau. \end{cases}$$

2. GAUSSIAN PHASE AVERAGING

For the purpose of simulating the field defined in the previous section, we wish to derive an expression for the observed signal at time $t = 2\tau$,

$$S(2\tau) = \langle e^{i\phi(t)} \rangle.$$

Proposition 1. *If ϕ is a zero-mean Gaussian random variable, then*

$$\langle e^{i\phi} \rangle = \exp \left[-\frac{1}{2} \text{Var}(\phi) \right]$$

Proof. One can check directly by evaluating the expectation $\langle e^{i\phi} \rangle$ with $\phi \sim \mathcal{N}(0, v)$ $\square$

Since $\mathfrak{B}(t)$ is Gaussian and ϕ is a linear functional of $\mathfrak{B}(t)$, ϕ is Gaussian. Therefore,

$$\begin{aligned} S(2\tau) &= \langle e^{i\phi} \rangle \\ &= \exp \left[-\frac{1}{2} \gamma^2 \int_0^{2\tau} \int_0^{2\tau} y(t) y(s) \langle \mathfrak{B}(t) \mathfrak{B}(s) \rangle dt ds \right] \\ &= \exp \left[-\frac{1}{2} \gamma^2 \Delta^2 \int_0^{2\tau} \int_0^{2\tau} y(t) y(s) e^{-|t-s|/\tau_c} dt ds \right]. \end{aligned}$$

3. EVALUATING THE DOUBLE INTEGRAL

Let

$$I(\tau) = \int_0^{2\tau} \int_0^{2\tau} y(t) y(s) e^{-|t-s|/\tau_c} dt ds.$$

Then, define

$$\begin{aligned} I_{11} &= \int_0^\tau \int_0^\tau e^{-|t-s|/\tau_c} dt ds, \\ I_{22} &= \int_\tau^{2\tau} \int_\tau^{2\tau} e^{-|t-s|/\tau_c} dt ds, \\ I_{12} &= \int_0^\tau \int_\tau^{2\tau} e^{-|t-s|/\tau_c} dt ds, \\ I_{21} &= \int_\tau^{2\tau} \int_0^\tau e^{-|t-s|/\tau_c} dt ds, \end{aligned}$$

so that $I(\tau) = I_{11} + I_{22} - I_{12} - I_{21}$. One quickly sees that $I_{11} = I_{22}$ and $I_{12} = I_{21}$, yielding

$$I(\tau) = 2I_{11} - 2I_{12}.$$

For I_{11} we have

$$\begin{aligned} I_{11} &= \int_0^\tau \int_0^\tau e^{-|t-s|/\tau_c} dt ds \\ &= 2 \int_0^\tau \int_0^t e^{-(t-s)/\tau_c} dt ds \\ &= 2\tau_c \int_0^\tau (1 - e^{-t/\tau_c}) dt \\ &= 2\tau_c [t + \tau_c e^{-t/\tau_c}]_0^\tau \\ &= 2\tau_c \tau - 2\tau_c^2 (1 - e^{-\tau/\tau_c}). \end{aligned}$$

Likewise, for I_{12} ,

$$\begin{aligned} I_{12} &= \int_0^\tau \int_\tau^{2\tau} e^{-|t-s|/\tau_c} dt ds \\ &= \int_0^\tau \int_\tau^{2\tau} e^{-(s-t)/\tau_c} dt ds \\ &= \left(\int_0^\tau e^{t/\tau_c} dt \right) \left(\int_\tau^{2\tau} e^{-s/\tau_c} ds \right) \\ &= \tau_c^2 (e^{\tau/\tau_c} - 1) (e^{-\tau/\tau_c} - e^{-2\tau/\tau_c}) \\ &= \tau_c^2 (1 - e^{-\tau/\tau_c})^2. \end{aligned}$$

Therefore,

$$I(\tau) = 4\tau_c \tau - 6\tau_c^2 + 8\tau_c^2 e^{-\tau/\tau_c} - 2\tau_c^2 e^{-2\tau/\tau_c},$$

giving us

$$S(2\tau) = \exp \left[-\frac{1}{2} \gamma^2 \Delta^2 (4\tau_c \tau - 6\tau_c^2 + 8\tau_c^2 e^{-\tau/\tau_c} - 2\tau_c^2 e^{-2\tau/\tau_c}) \right].$$